

Running the MBE seed on Colab — instructions for coauthors

Goal: produce the real 7–8B "KV Compression Cards" that go into Section 10 / the leaderboard of our survey. Right now the paper only has a 0.5B CPU proof-of-concept; this gives us the model-scale numbers. It takes ~20–40 minutes on a **free** Colab GPU. You need: a Google account. (Optional: a Hugging Face account — only if you use a gated model like Llama, or want to upload the result yourself.)

Step 1 — Open the notebook

Click the badge in the repo README, or open this link:

https://colab.research.google.com/github/rohithreddybc/kv-cache-compression-mbe/blob/main/colab/MBE_seed_run.ipynb

Step 2 — Turn on the GPU (important)

In Colab: **Runtime** → **Change runtime type** → **Hardware accelerator** → **T4 GPU** → **Save**.

(The free T4 has 16 GB, enough — the 7–8B model is loaded in 4-bit.)

Step 3 — Run the baseline cells

We want the built-in baselines (KIVI quantization, StreamingLLM, H2O), so use **Option B**.

1. Run **cell 1** ("Install + check GPU"). It installs packages and clones the repo (~2 min). You should see a line like `NVIDIA T4 ... 15360 MiB`. If it says no GPU, go back to Step 2.
2. **Skip the "Option A" cell** (that's for evaluating a brand-new method).
3. Run the **"Option B — baselines"** cell. Default model is Qwen/Qwen2.5-7B-Instruct (open, no access needed). This is the one to run for our numbers. It prints, per method and budget, a line like:

```
kivi-4bit      budget=25.0%  -> passkey acc 0.950
streaming     budget=25.0%  -> passkey acc 0.000
...
```

4. Run the **"Show your card"** cell. It prints the result JSON and saves it to `mbe-protocol/cards/seed_multimethod_qwen2.5-7b-instruct.json`.

Step 4 — Send the result back

Two options, either is fine:

- **Easiest:** in Colab's left sidebar open the **Files** panel → `mbe-protocol/cards/` → right-click the new `seed_*.json` → **Download**, and email/Slack it to the corresponding author (Rohith). He'll commit it to the repo and the leaderboard.
- **Or upload it yourself** (last cell): it asks you to log in with a Hugging Face **write** token (<https://huggingface.co/settings/tokens>). It uploads to **your own** HF dataset namespace; then open an issue/PR on the GitHub repo with the link so it's added to the central leaderboard.

That's it. **Cells 1 → Option B → Show card → send the JSON.**

Optional: run a second model

To strengthen the table, re-run the **Option B** cell after changing the model line to another from the dropdown (e.g. `meta-llama/Llama-3.1-8B-Instruct`) and run "Show card" again. **Llama is gated:** you must first request access on its HF model page and run `from huggingface_hub import login; login()` with your token in a cell. Qwen needs none.

Troubleshooting

- **Out-of-memory (CUDA OOM):** set `MBE_SKIP_H2O = '1'` in the Option B cell (H2O needs the most memory), or switch the model to Qwen/Qwen2.5-3B-Instruct.
- **"No GPU" / very slow:** you didn't enable the T4 (Step 2), or Colab gave you a CPU runtime — reconnect and retry.
- **Disconnects after idle:** Colab free tier times out if idle; keep the tab active, or just rerun — it's deterministic (fixed seed).
- **Want more samples (more stable numbers):** raise `MBE_N` (e.g. `'40'`); it's slower.

What the numbers mean

Each method is evaluated at the **same KV-memory budget** (50/25/12.5%) on a passkey retrieval task. Quantization keeps every token (usually near-lossless); eviction methods

drop tokens (and on this single-needle task can miss the needle). The point of MBE is exactly this apples-to-apples, fixed-budget comparison — see Section 10 of the paper.

Links

- Notebook: `colab/MBE_seed_run.ipynb` (in this repo)
- Repo: <https://github.com/rohithreddybc/kv-cache-compression-mbe>
- Dataset: <https://huggingface.co/datasets/Rohithreddybc/kv-cache-compression-mbe>
- Leaderboard: <https://huggingface.co/spaces/Rohithreddybc/kv-cache-compression-leaderboard>